

CORNELL MASTER BEEKEEPING - PROGRAM SYLLABUS

1 - HONEY BEE EVOLUTION, BIOLOGY, AND BEHAVIOR

Module 1 Introduction: The Evolution of the Honey Bee Colony as a Superorganism

The Honey Bee Colony as a Superorganism
Other Superorganisms
What Does It Mean to Be a Honey Bee?
Distinguishing Honey Bees From Other Insects
Distinguishing Honey Bees From Other Insects
The Evolution of Honey Bees
Eusociality: How Honey Bees Are Truly Social
How Did Eusocial Societies With Distinct Castes Evolve?
Kin Selection
Polyandry
The Importance of Wild Bees
Conserving Wild Bees
The Evolution of the Honey Bee Colony as a Superorganism
Module 1 Discussion and Questions
Module Wrap-up: The Evolution of the Honey Bee Colony as a Superorganism

Module 2 Introduction: Honey Bee Reproduction

Passing on Your Genes
The Molecules of Bee Life
Mating
Reproductive Systems
Honey Bee Reproduction
Module 2 Discussion and Questions
Module Wrap-up: Honey Bee Reproduction

Module 3 Introduction: Honey Bee Development

The Stages of Development
The Stages of Development Timeline
How Sex Is Determined
The Complementary Sex Determination Gene
Diploid Male Occurrences
How Caste Is Determined
Molecules and Food Involved in Caste Determination
Intercastes
How the Worker's Role Transitions Throughout Its Life
Worker Bee Ages and Tasks
Molecules Involved in Worker Polyethism
Social Environment Influence on Worker Tasks
Bee Lifespan
Honey Bee Development
Module 3 Discussion and Questions

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Module Wrap-up: Honey Bee Development

Module 4 Introduction: How Honey Bees Communicate

Pheromone Communication

Honey Bee Dances — The Waggle Dance

Honey Bee Dances — The Shaking Signal

Honey Bee Dances — The Tremble Dance and “Stop” Signal

How Are Dances Perceived in the Dark?

Honey Bee Communication

Module 4 Discussion and Questions

Module Wrap-up: How Honey Bees Communicate

Module 5 Introduction: Honey Bee Biology

The Nervous System

Sensory Systems: Mechanoreception

Sensory Systems: Olfaction (Smell)

Sensory Systems: Gustation (Taste)

Sensory Systems: Vision

How Understanding Honey Bee Vision Can Make You a Better Beekeeper

The Muscular System: Flight

The Respiratory System

The Circulatory System

The Digestive System

Honey Bee Biology

The Amazing Fat Body

Module 5 Discussion and Questions

Module Wrap-up: Honey Bee Biology

Module 6 Introduction: How Honey Bees Construct and Maintain Their Nest

The Building Blocks of a Nest

How a Nest of a Honey Bee Colony Is Organized — Location and Foraging Area

How a Nest of a Honey Bee Colony Is Organized — Bee Space

How a Nest of a Honey Bee Colony Is Organized — Nest Characteristics

Hive Styles

Temperature Control in the Nest

Temperature Control in the Nest — Heating

Temperature Control in the Nest — Cooling

Nest Construction and Maintenance

Module 6 Discussion and Questions

Course Project: Creating an Outline for a Presentation

Module Wrap-up: How Honey Bees Construct and Maintain Their Nests

2 - MANAGING PESTS & DISEASES

Module 1 Introduction: Honey Bee Immunity

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Individual-Level Defenses

Attacking Pathogens Inside the Body: the Honey Bee Immune System

Colony-Level Defenses: Social Immunity

Propolizing the Nest

The Microbiome

Beyond Immunity — Beekeeper Intervention

Honey Bee Immunity

Module Wrap-up: Honey Bee Immunity

Module 2 Introduction: Insect Pests

Wax Moth Biology

Wax Moth Identification in the Colony

Wax Moth Prevention

Wax Moth Treatment

Small Hive Beetle Biology

Small Hive Beetle Effects

Small Hive Beetle Appearance and Identification

Small Hive Beetle Prevention

Small Hive Beetle Treatment

Yellowjackets

Hornets

Insect Pests

Module Wrap-up: Insect Pests

Module 3 Introduction: Fungal Enemies — Nosema

Nosema Biology

Nosema Symptoms

How Can I Tell If My Colony Is Infected?

Prevalence and Seasonality

The Interaction Between Nosema and Pesticides

Nosema Prevention

Nosema Management

Fumagillin Is Not a Reliably Effective Treatment

Nosema

Module 3

Module Wrap-up: Fungal Enemies — Nosema

Module 4 Introduction: Varroa Mites

The Rise of Varroa

Varroa's Entry Into the U.S.

Life Cycle of Varroa

Varroa's Impact on Bee Health — Feeding

Varroa's Impact on Bee Health — Transmitting Viruses

Varroa's Impact on Bee Health — Parasitic Mite Syndrome

Varroa's Impact on Bee Health — Colony Death

A Leading Cause of Colony Losses

IPM for Varroa

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Monitoring Overview
How to Monitor
Unreliable Techniques for Monitoring Mites
Treatment Threshold
The Importance of Hygienic Behavior
How Is Hygienic Behavior Tested in Colonies?
Genetic Stocks for Varroa Resistance
Genetic Stocks for Varroa Resistance Overview
Genetic Stocks for Varroa Resistance — Varroa Sensitive Hygienic
Genetic Stocks for Varroa Resistance — Minnesota Hygienic
Genetic Stocks for Varroa Resistance — Purdue Ankle Biter
Genetic Stocks for Varroa Resistance — Russian
Genetic Stocks for Varroa Resistance — Saskatraz
Genetic Stocks for Varroa Resistance — Grooming Behavior
How to Introduce New Stock and Select for Varroa Resistance
Breeding for Resistance in Your Own Operation
Cultural Methods for Varroa Management Overview
Cultural Methods for Varroa Management — Drone Frame
Cultural Methods for Varroa Management — Screened Bottom Board
Cultural Methods for Varroa Management — Small Hives
Cultural Methods for Varroa Management — Colony Spacing
Cultural Methods for Varroa Management — Brood Interruption
Ineffective Cultural Methods for Varroa Management
Cultural Management Practices and Wild Honey Bee Colonies
Chemical Controls for Varroa Management Overview
Chemical Controls for Varroa Management
Chemical Controls for Varroa Management — Mite Away Quick Strips
One-Page Guide to Varroa Treatments
Safety and Biosecurity of Chemical Controls
The Controversy Surrounding Some Synthetic Chemical Treatments
Varied Management Practices
Developing Your Own Varroa Management Plan
Mite Management Calendars
Varroa Mites
Module 4
Alternative Varroa Management
Module Wrap-up: Varroa Mites

Module 5 Introduction: Viruses

An Overview of Viruses
Sacbrood Virus
Deformed Wing Virus
Deformed Wing Virus Transmission and Symptoms
Black Queen Cell Virus
Acute Bee Paralysis Virus Complex
Chronic Bee Paralysis
Other Viruses

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Managing Viruses

Viruses

Module 5

Module Wrap-up: Viruses

Module 6 Introduction: Other Honey Bee Parasites

History of Tracheal Mites

Tracheal Mites Biology

Tracheal Mites Symptoms

Diagnosing Tracheal Mites

Treating Tracheal Mites

Zombees: Honey Bees Parasitized by Phorid Flies

Honey Bee Parasites

A New Parasite on the Horizon? Tropilaelaps mites

Module Wrap-up: Other Honey Bee Parasites

Module 7 Introduction: Brood Diseases

Chalkbrood

American Foulbrood: An Overview

American Foulbrood Symptoms

Inspecting for an AFB Infection

Tests to Diagnose an AFB Infection

You Have a Positive AFB Result; Now What?

The History of Antibiotics

New Antibiotic Regulations

How Does Resistance Develop?

Preventing AFB Infection in Your Colonies

European Foulbrood Overview

European Foulbrood Symptoms

Diagnosing and Treating EFB

Idiopathic Brood Disease Syndrome

Brood Diseases

Module 7

Module Wrap-up: Brood Diseases

Module 8 Introduction: Bee Health — The Big Picture

Honey Bee Colony Declines

Colony Collapse Disorder

Colony Collapse Disorder Investigation

Honey Bee Health in History

Tying It All Together

Honey Bee Symptom Chart

Pathogen Spillover to Wild Bees

How Do Wild Bees Catch These Infections?

Bee Health: The Big Picture

What Does It Mean To Be a Master Beekeeper?

Course Project — Fighting the Mite in Bee Yards

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Module Wrap-up: Bee Health — The Big Picture

3 - THE SCIENCE & ART OF BEEKEEPING

Module 1 Introduction: Safety and Biosecurity in the Bee Yard

- Preventing Stings
- Bee Sting Reactions
- Hygienic Management Practices
- Robbing
- How to Prevent Robbing
- How to Stop Robbing
- Safety and Biosecurity in the Bee Yard
- Minimizing Competition Between Your Hives and Native Bee Species
- Module 1 Discussion and Questions
- Module Wrap-up: Safety and Biosecurity in the Bee Yard

Module 2 Introduction: Inspecting the Colony

- Preparing to Inspect Your Hives
- What to Look For When Inspecting
- Walk Through a Hive Inspection
- Outside the Hive
- What Can We Learn From Bees at the Entrance?
- Inspecting During Less-Than-Perfect Conditions
- Inspecting Honey Supers
- Starting Your Inspection of the Brood Nest
- What to Look For When Inspecting the Brood Nest
- Inspection Tips
- Beekeeping Calendar
- A Year in the Bee Yard
- Colony Assessments
- Keeping Good Records
- Record Sheets
- Record Keeping Tips
- Tech-Enhanced Beekeeping
- AI for Beekeepers
- Inspecting the Colony
- Module 2 Discussion and Questions
- Module Wrap-up: Inspecting the Colony

Module 3 Introduction: Honey Bee Nutrition

- Honey Bee Nutrition Overview
- Nutritional Requirements — The How and Why of Bee Food
- Natural Forage

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Valuable Pollen Plants
Learn the Blooms in Your Area
When to Feed Carbohydrates
How to Feed Carbohydrates
Dry Feeding Carbohydrates in the Winter
Feeding Protein
Protein Feeding Options
Honey Bee Nutrition
Langstroth Hives: Not the Only Hive Design Available
Module Wrap-up: Honey Bee Nutrition

Module 4 Introduction: Techniques in Managing Colony Population

Annual Population Cycle of a Honey Bee Colony
Reasons for Splitting Colonies
How To Make a Proper Split
Splitting Colonies
Alternate Ways to Split a Colony
Nucleus Colonies
How to Make a Nuc
The Composition of Constructed Nucs
Wintering Nucs
Merging Colonies Overview
Merging Colonies — The Newspaper Method
Merging Colonies
Merging a Nuc With an 8 to 10 Frame Colony
Techniques in Managing Colony Population
Module 4 Discussion and Questions
Module Wrap-up: Techniques in Managing Colony Population

Module 5 Introduction: Swarming

Colony-Level Reproduction — Swarming
Swarming Timing and Initiation
The Preparation and Process of Swarming
The Swarm Event
Swarming or Not?
Swarm Prevention
Swarm Control
Ineffective Practices for Swarm Control
My Colony Swarmed! Now What?
Should I Let My Colonies Swarm?
Catching and Baiting Swarms
Swarming
Module 5 Discussion and Questions
Module Wrap-up: Swarming

Module 6 Introduction: Queen Management

Queen Status

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Signs of Queenlessness
Considerations for Requeening a Queenless Colony
Tip for Introducing a New Queen
Requeening to Improve Health and Productivity
Managing a Two-Queen Colony
How To Make a Two-Queen Colony
Using Local Queens
What Are The Benefits of Using Local Bees?
Queen Management
Why Are Queens Failing Faster Than Before?
Module Wrap-up: Queen Management

Module 7 Introduction: Variation Within *Apis mellifera*

Honey Bee Subspecies and Lineages
Apis Mellifera in North America
Africanized Honey Bees
Variation Within *Apis mellifera*
Module 7 Discussion and Questions
Module Wrap-up: Variation Within *Apis Mellifera*
Module Introduction: Queen Rearing With the Doolittle Method
The Doolittle Method
Biological Considerations
Queen Grafting
Queen Grafting Example
Considerations When Placing Your Graft in Cell Builders
Cell Builders — The Cell Starter
Cell Builders — Cell Finisher
Mating Yards
Queen Rearing Schedule
Queen Rearing With the Doolittle Method
Module Wrap-up: Queen Rearing With the Doolittle Method

Module 8 Introduction: Breeding for the Traits You Desire

Breeding Bees
Genetic Diversity
Inbreeding
Overview of Breeding Bees
The Role of Drones
Controlled Mating vs. Open Mating
Evaluating and Comparing Colonies' Traits
Evaluating Multiple Traits Across Multiple Yards
Breeding for the Traits You Desire
Modules 8 and 9 Discussion and Questions
Course Project — Plan Your Upcoming Bee Year
Module Wrap-up: Breeding for the Traits You Desire

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THE REWARDS AND CONTRIBUTIONS OF BEEKEEPING

Module 1 Introduction: Honey

- From Nectar to Liquid Gold
- Medicinal Benefits of Honey
- Manuka Honey: The Most Antibacterial Honey
- Composition of Honey
- Changes in Honey Over Time — Crystallization
- Changes in Honey Over Time — Fermentation
- Processing Liquid Honey
- Components and Design of a Honey House
- Types of Extractors
- Other Honey House Considerations
- Virtual Field Trip to Honey Houses
- Selling Local Honey as a Small-Scale Beekeeper
- Packaging and Labeling the Product
- Selling Honey as a Sideliner or Commercial Beekeeper
- Selling Honey as a Sideliner or Commercial Beekeeper — Influences on Market Price
- Selling Honey as a Sideliner or Commercial Beekeeper — Honey Grades and Standards
- Honey
- The Rewards and Contributions of Beekeeping
- Module Wrap-up: Honey

Module 2 Introduction: Additional Hive Products

- Beyond Honey
- Rendering Beeswax
- Methods of Rendering Beeswax for Hobby Operations
- Processing Beeswax Example
- The Many Uses of Beeswax
- The Many Uses of Beeswax — Making Candles
- Pollen
- Medicinal Properties of Pollen
- Harvesting Pollen
- Propolis
- Harvesting Propolis
- Preparing Propolis
- Royal Jelly
- Venom
- Mead
- Making Mead — The Basics
- Additional Hive Products
- Additional Hive Products
- Module Wrap-up: Additional Hive Products

Module 3 Introduction: Pollination Services

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Pollination: The Birds and the Bees
Preparing Colonies for Pollination
Hive Stocking Rates
Renting Hives — Costs
Renting Hives — Benefits
Creating a Pollination Contract
Preparing to Transporting Hives
Transporting Hives — Loading the Truck
Transporting Hives — Travel Considerations
Pollination Services
The Future of Pollination
Module Wrap-up: Pollination Services

Module 4 Introduction: Confidence in Communicating About Pesticides

An Overview of Pesticides — What's in a Pesticide?
Pesticides in the Environment and in the Hive
Measuring Harm to Honey Bees
An Overview of Insecticides
Insecticides
A Special Class of Insecticides: Neonicotinoids
Toxicity of Neonicotinoids
Neonicotinoid Encounters Cause Sublethal and Chronic Effects
Herbicides
Fungicides
Fungicides — Synergistic Effects
Reducing Bees' Exposure to Pesticides
Confidence in Communicating About Pesticides
Interpreting Pesticide Results
Confidence in Communicating About Pesticides
Module Wrap-up: Confidence in Communicating About Pesticides

Module 5 Introduction: Turning Bees Into Business

Considerations When Growing Your Operation
Considerations When Growing Your Operation — Record Keeping
Inventory and Financial Analysis Handbook
Building Your Business
Mindset and Management Changes
Deciding on Business Structure
Programs and Resources to Help Beekeepers Overview
Programs to Help With Colony Health
Programs to Help With Your Beekeeping Business
Turning Bees Into Business
Module Wrap-up: Turning Bees Into Business

Module 6 Introduction: Being a Leader in Your Community

Mentoring
A Sample Mentoring Schedule

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Community Service
Participating in Research
Delivering Effective Presentations
Turning Bees Into Business and Being a Leader in Your Community
Being a Leader in Your Community
Module Wrap-up: Being a Leader in Your Community

Module 7 Introduction: Keeping Up With Science

Where to Find New Information
Popular Science Articles
Obtaining Reputable Scientific Articles
Developing Skills in Reading Scientific Articles — Understanding Layout
Developing Skills in Reading Scientific Articles — Interpreting Papers
Reading and Interpreting Scientific Article Example
Distinguishing Science From Pseudoscience
Exploring Your Curiosity: How to Design Your Own Study
Interpreting a Scientific Paper
Keeping Up With Science
Final Project